

Module - 6 : Hypothesis Testing II

Test of Significance for small samples :-

Test of Significance discussed in the previous section hold good only for large samples, i.e. only when the size of the sample $n \geq 30$. When the sample is small i.e. $n < 30$, the sampling distributions of many statistics are not normal, even though the parent populations may be normal. Moreover, the assumption of near equality of population parameters and the corresponding sample statistics will not be justified for small samples. Consequently, we have to develop entirely different tests of significance that are applicable to small samples.

Hence, in the study of small sample, the test statistics will change. In all exact sample tests, we assume, that population from which samples are drawn are normal.

Student's t-distribution:

William Sealy Gosset

t-distribution is defined by the mathematician W.S. Gosset (1908) who wrote under the pen name of Student, and later on developed and extended by R.A. Fisher (1926).

Student's t-Distribution:

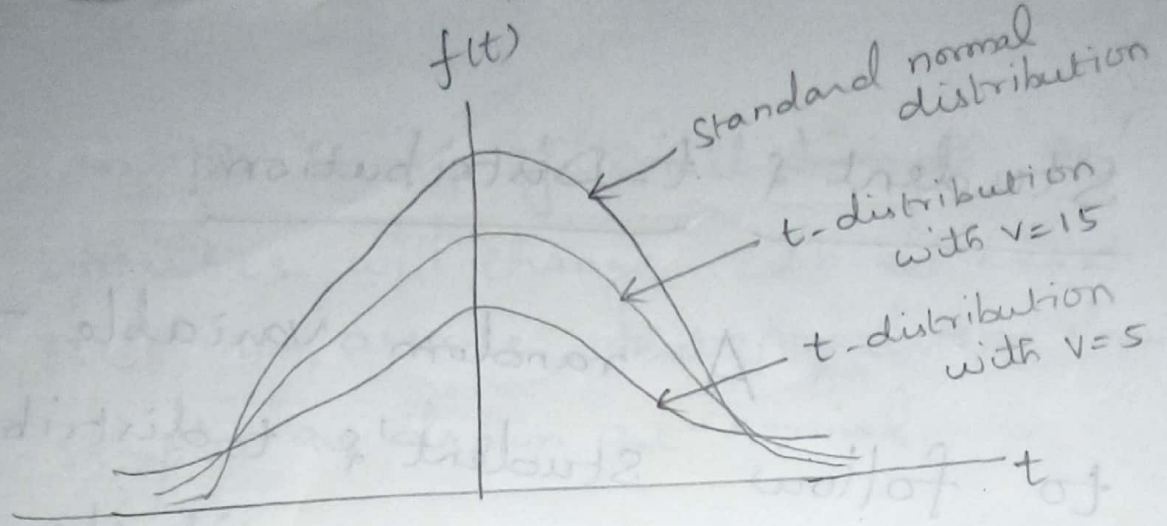
A random variable T is said to follow Student's t -distribution or simply t -distribution, if its probability density function is given by

$$f(t) = \frac{1}{\sqrt{v} \beta(\frac{v}{2}, \frac{1}{2})} \left(1 + \frac{t^2}{v}\right)^{-\frac{(v+1)}{2}}, \quad -\infty < t < \infty$$

where v is called the number of degrees of freedom of the t -distribution.

Properties of t -distribution:

1. The probability curve of the t -distribution is similar to the standard normal curve and is symmetric about $t=0$, bell-shaped and asymptotic to the t -axis as shown below.



2. For sufficiently large value of n , the t -distribution tends to the standard normal distribution. $v \rightarrow \infty$
3. The mean of the t -distribution is zero.
4. The variance of the t -distribution is $\frac{v}{v-2}$ if $v > 2$
5. The variance is always greater than 1 although it is close to 1 when $v \rightarrow \infty$.

Applications (uses) of t-distribution:

The t-distribution is used to test the significance of the difference between

1. The mean of a small sample and the mean of the population.
2. The means of two small samples.
3. The observed sample correlation coefficient and sample regression coefficient.
4. The Observed partial and multiple correlation coefficients.

Assumptions for Student's t test

The following assumptions are made in Student's t test.

1. The parent population from which sample is drawn is normal.
2. The sample observations are independent, that is, sample is random.
3. The population standard deviation σ is unknown.
4. Sample size $n < 30$.

Degrees of freedom:

The number of degrees of freedom, usually denoted by the Greek alphabet ν , can be interpreted as the number of useful bits of information generated by a sample of given size for estimating a population parameter.

Suppose we wish to find the mean of a sample with observations $x_1, x_2, \dots, x_n$. We have to use all the 'n' values taken by the variable with full freedom (ie without putting any constraint or restriction on them) for computing $\bar{x}$. Hence $\bar{x}$ is said to have n degrees of freedom.

Suppose we wish to further compute the S.D. 's' of this sample using the formula $s^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$. Though we use the n values

$x_1 - \bar{x}, x_2 - \bar{x} \dots x_n - \bar{x}$ for this computation, they do not have 'n' degrees of freedom, as they depend on $\bar{x}$ which has been already calculated and fixed. Since there is one restriction regarding the value of $\bar{x}$, 's' is said to have $(n-1)$ degrees of freedom.

If we compute another statistic of the sample based on $\bar{x}$ and s , then that statistic will be assumed to have $(n-2)$ degrees of freedom, and so on.

Thus, the number of independent variates used to compute the test statistic is known as the number of degrees of freedom of that statistic. In general, the number of degrees of freedom is given by $v = n - k$, where n is the number of observations in the sample and k is the number of constraints imposed on

them or k is the number of values that have been found out and specified by prior calculations.

Critical value of t :

The critical value or significant value of t at level of significance α and degrees of freedom v , for two tailed test, is given by

$$P\{|t| > t_v(\alpha)\} = \alpha$$

$$P(|t| \leq t_v(\alpha)) = 1 - \alpha.$$



The significant value of t at level of significance ' α ' for a single-tailed test can be got from those of two-tailed test by referring to the values at ' 2α '. critical value t_{α} are given in the tables, called t -table.

Student's t-distribution is used when Sample size is ≤ 30 and the population standard deviation is unknown.

t-statistic is defined as

$$t = \frac{\bar{x} - \mu}{S/\sqrt{n}} \quad \text{where } S = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$$

$\bar{x}$ is the mean of sample, μ is population mean. S is the standard deviation of population and n is sample size.

If the standard deviation of the sample ' s ' is given then t-statistic is defined as

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}}$$

Note: The relation between s and S is $ns^2 = (n-1)S^2$.

Test - 1:

Test of Significance of the difference between Sample mean and population mean.
(or) t-test of Significance for single mean.

Under the null hypothesis H_0 : The Sample has been drawn from population with mean μ or there is no significant difference between the Sample mean $\bar{x}$ and the population mean μ , we use the statistics

$$t = \frac{\bar{x} - \mu}{S/\sqrt{n}} \quad (\text{or}) \quad \frac{\bar{x} - \mu}{s/\sqrt{n-1}}$$

Where $\bar{x}$ is mean of the sample and

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \text{ with degrees of freedom } (n-1).$$

At given level of Significance α , and degrees of freedom $(n-1)$, we refer to t-table value t_α (two-tailed or one-tailed).

If calculated t value is such that
 $|t| < t_\alpha$, the null hypothesis is accepted
If $|t| > t_\alpha$, H_0 is rejected.

Confidence or fiducial limits for population mean μ :

If t_α is the value of t at level of significance α at $(n-1)$ degrees of freedom then,

$$\left| \frac{\bar{x} - \mu}{S/\sqrt{n}} \right| < t_\alpha \text{ for acceptance of } H_0.$$

$$\bar{x} - t_\alpha \frac{S}{\sqrt{n}} < \mu < \bar{x} + t_\alpha \frac{S}{\sqrt{n}}$$

95%. Confidence limits (level of significance 5%)

$$\text{are } \bar{x} \pm t_{0.05} \frac{S}{\sqrt{n}}$$

99%. Confidence limits (level of significance 1%)

$$\text{are } \bar{x} \pm t_{0.01} \frac{S}{\sqrt{n}}$$

① A sample of 26 bulbs gives a mean life of 990 hours with a S.D of 20 hours. The manufacturer claims that the mean life of bulbs is 1000 hours. Is the sample not upto the standard.

Solution:

Here Sample size $n = 26 < 30$

$\therefore$ The sample is small sample.

Also given, Sample mean $\bar{x} = 990$

population mean $\mu = 1000$

Standard deviation of the sample $s = 20$

Degrees of freedom $= n - 1 = 26 - 1 = 25$

Null hypothesis H_0 : The sample is upto the standard

Alternative hypothesis H_1 : $\mu < 1000$ (the sample is below standard) (left tailed test)

The test statistic is $t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}}$

$$t = \frac{990 - 1000}{\frac{20}{\sqrt{25}}}$$

$$t = -2.5$$

$$|t| = 2.5$$

Tabulated 't' at 5% level with 25 degrees of freedom for left-tailed test is 1.708

Since calculated $t >$ tabulated t ,
we reject H_0 .

i.e. we reject the null hypothesis.

$\therefore$ The sample is not upto the standard.

② The mean weekly sales of Soap bars in departmental stores was 146.3 bars per store. After an advertising campaign the mean weekly sales in 22 stores for a typical week increased to 153.7 and showed a S.D. of 17.2. Was the advertising campaign successful.

Solution:

Here the number of stores $n = 22 < 30$
i.e. the sample size < 30 .

$\therefore$ The small is small.

Also given Sample mean $\bar{x} = 153.7$

Population mean $\mu = 146.3$

Sample S.D. $s = 17.2$

Degrees of freedom $= n - 1 = 22 - 1 = 21$

Null Hypothesis H_0 : The advertising campaign was not successful.

Alternative hypothesis $H_1: \mu > 146.3$

(Right tailed test)

The test statistic is

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n-1}}$$

$$= \frac{153.7 - 146.3}{\frac{17.2}{\sqrt{22-1}}}$$

$$= 1.97$$

Calculated value of $t = 1.97$

Tabulated value of t at 5% level with 21 degrees of freedom for single tailed test is 1.721.

Since calculated $t >$ tabulated t , we reject H_0 .

i.e. we reject the assumption that the advertising Campaign was not successful. i.e. advertising Campaign was successful.

③ The mean life time of a sample of 25 fluorescent light bulbs produced by a company is computed to be 1570 hours with a S.D. of 120 hours. The company claims that the average life of the bulbs produced by the company is 1600 hours. Using the level of significance of 0.05. Is the claim acceptable?

Solution:

Sample size $n = 25$

Sample mean $\bar{x} = 1570$

Population mean $\mu = 1600$

Sample S.D $\sigma = 120$

Degrees of freedom $= n - 1 = 24$

Null hypothesis H_0 : The claim is acceptable.
i.e. $\mu = 1600$ hrs.

Alternative hypothesis H_1 : $\mu \neq 1600$ hrs.

The test statistic is $t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}}$

$$= \frac{1570 - 1600}{\frac{120}{\sqrt{24}}}$$

$$t = -1.22$$

$$|t| = 1.22$$

Calculated $t = 1.22$

The tabulated value of t at 5% level with 24 degrees of freedom for two tailed test is 2.064.

Since the Calculated value of $t <$ tabulated value of t .

we accept the null hypothesis H_0 .
i.e. the claim that the average life of the bulbs produced by the company is 1600 hours is acceptable.

S.D of the sample is not given directly
④ A random Sample of size 16 has 53 as mean. The sum of squares of the deviation from mean is 135. Can this sample be regarded as taken from the population having 56 as mean? obtain 95% and 99% Confidence limits of the mean of the population.

Solution:

Given $\bar{x} = 53$, $\mu = 56$, $n = 16$,

$$\sum (x - \bar{x})^2 = 135$$

$$S = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}} = \sqrt{\frac{135}{15}} = 3$$
$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n}} = \sqrt{\frac{135}{16}} = 2.9047$$

Null hypothesis H_0 : There is no significant difference between the sample mean and hypothetical population mean i.e. $\mu = 56$.

Alternative hypothesis $H_1: \mu \neq 56$

(Two tailed test)

Under H_0 , test statistic is

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}}$$
$$= \frac{53 - 56}{\frac{2.9047}{\sqrt{15}}}$$

$$= \frac{-3}{0.7494}$$

$$= -4$$

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$$= \frac{53 - 56}{3/\sqrt{16}} = -4$$

$$|t| = 4$$

degrees of freedom $= 16 - 1 = 15$.

Since $|t| = 4 > t_{0.05} = 2.131$.

ie the calculated value of t is more than the tabulated value, the null hypothesis is rejected. Hence, the sample mean has not come from a population having 56 as mean.

95% Confidence limits of the population

$$\text{mean} = \bar{x} \pm \frac{S}{\sqrt{n}} t_{0.05}$$

$$= 53 \pm \frac{3}{\sqrt{16}} (2.13)$$

$$= (51.4025, 54.5975)$$

99% Confidence limits of the population

$$\text{mean} = \bar{x} \pm \frac{S}{\sqrt{n}} t_{0.01}$$

$$= 53 \pm \frac{3}{\sqrt{16}} (2.95)$$

$$= (50.7875, 55.2125)$$

⑤ The nine items of a sample had the following values 45, 47, 50, 52, 48, 47, 49, 53, 51. Does the mean of the nine items differ significantly from the assumed population mean 47.5.

Solution:

First we have to find sample mean and sample standard deviation as follows.

x	$x - \bar{x}$	$(x - \bar{x})^2$
45	-4.11	16.89
47	-2.11	4.45
50	0.89	0.79
52	2.89	8.35
48	-1.11	1.23
47	-2.11	4.45
49	-0.11	0.012
53	3.89	15.13
51	1.89	3.57
<u>442</u>		<u>54.872</u>

$$\text{Mean } \bar{x} = \frac{\sum x}{n} = \frac{442}{9} = 49.11$$

$$S^2 = \frac{1}{n-1} \sum (x - \bar{x})^2$$

$$= \frac{1}{8} \times 54.872$$

$$S^2 = 6.859$$

$$S = 2.62$$

$$S^2 = \frac{1}{n} \sum (x - \bar{x})^2$$

$$= \frac{1}{9} \times 54.872$$

$$= 6.09688$$

$$= 2.4692$$

Null Hypothesis H_0 : The sample mean $\bar{x}$ does not differ significantly from the population mean μ .

Alternative hypothesis H_1 : $\mu \neq 47.5$
(two tailed test)

Test statistic is

$$t = \frac{\bar{x} - \mu}{S/\sqrt{n}}$$

$$t = \frac{\bar{x} - \mu}{S/\sqrt{n}}$$

$$= \frac{49.11 - 47.5}{\frac{2.4692}{\sqrt{9}}} = \frac{1.61}{0.8197333} = 1.8442$$

$$= \frac{49.11 - 47.5}{\frac{2.62}{\sqrt{9}}} = 1.84$$

Calculated $t = 1.84$

degrees of freedom $= n - 1 = 9 - 1 = 8$

Tabulated value of t for 8 degrees of freedom at 5% level of significance is 2.31

Since calculated $t < \text{tabulated } t$,
we accept H_0 .

i.e. the two means $\bar{x}$ and μ do not
differ significantly.

- ⑥ The lifetime of electric bulbs for a
random sample of 10 from a large
Consignment gave the following data.

Item	1	2	3	4	5	6	7	8	9	10
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Life in hrs	4.2	4.6	3.9	4.1	5.2	3.8	3.9	4.3	4.4	5.6
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Can we accept the hypothesis that the average
lifetime of bulb is 4000 hrs?

Solution:

First we have to find the sample
mean and sample standard deviation
as follows.

x	$x - \bar{x}$	$(x - \bar{x})^2$
4.2	-0.2	0.04
4.6	0.2	0.04
3.9	-0.5	0.25
4.1	-0.3	0.09
5.2	0.8	0.64
3.8	-0.6	0.36
3.9	-0.5	0.25
4.3	-0.1	0.01
4.4	0	0
5.6	1.2	1.44
<u>44.0</u>		<u>3.12</u>

$$\bar{x} = \frac{\sum x}{n} = \frac{44}{10} = 4.4$$

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n}} = \sqrt{\frac{3.12}{10}} = 0.5586$$

$$S = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}} = 0.589$$

Null hypothesis H_0 : There is no significant difference in the sample mean and the population mean. i.e. $\mu = 4000$ hrs.

Alternative hypothesis $\mu \neq 4000\text{hrs}$
(Two tailed test)

Under H_0 , the test statistic is

$$t = \frac{\bar{x} - \mu}{S/\sqrt{n}}$$

$$= \frac{4.4 - 4}{\frac{0.589}{3}}$$

$$= \frac{.4}{0.1962} = 2.1482$$

$$t = \frac{\bar{x} - \mu}{S/\sqrt{n}}$$

$$= \frac{4.4 - 4}{\frac{0.589}{\sqrt{10}}} = 2.123$$

degrees of freedom $= 10 - 1 = 9$

Tabulated value of t for 9 degrees of freedom at 5% level of significance is 2.26.

Since Calculated $t <$ tabulated t ,
we accept H_0 .

i.e. the average lifetime of bulbs
could be 4000 hrs.